# Derived Categories

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## 1 The Derived Category

We want invariants for spaces (identifying homotopic spaces). But taking homology is not good enough, as there are non-homotopic spaces which have the same homology.

**Motto 1.1.** *"Complexes are good, homology of complexes is bad."*

We start with an abelian category $\mathcal{A}$, and construct a new category $\mathrm{Comp}(\mathcal{A})$ whose objects are complexes of objects in $\mathcal{A}$, and maps are chain maps (mind that we use cohomological notation).

$$A^\bullet \quad = \quad \cdots \xrightarrow{d^{i-2}} A^{i-1} \xrightarrow{d^{i-1}} A^i \xrightarrow{d^i} A^{i+1} \xrightarrow{d^{i+1}} \cdots$$

with $d^{i+1} \circ d^i = 0$. A morphism $f : A^\bullet \to B^\bullet$ is a collection of maps $f^i : A^i \to B^i$ such that $d^i \circ f^i = f^{i+1} \circ d^i$.

We say two chain maps $f^\bullet, g^\bullet : A^\bullet \to B^\bullet$ are *homotopic* if an only if there exists a map $h^\bullet$ such that

$$f^i - g^i = d^{i-1} \circ h^i + h^{i+1} \circ d^i.$$

One can immediately show that this makes "being homotopic" an equivalence relation.

**Definition 1.2.** Let $\mathcal{A}$ be an abelian category. We define the *homotopy category* $\mathbf{K}(\mathcal{A})$ to be the category whose objects are complexes in $\mathcal{A}$, and morphisms are chain maps modulo the homotopy equivalence relation.

**Proposition 1.3.** *Let $A, B$ be objects of $\mathcal{A}$, and $f : A \to B$ a morphism. Let $0 \to A \to I_A^\bullet$ and $0 \to B \to I_B^\bullet$ be injective resolutions of $A$ and $B$, respectively. Then the map $f : A \to B$ can be lifted to a map $f^\bullet : I_A^\bullet \to I_B^\bullet$ such that $H^0(f^\bullet) = f$ (i.e. $f^\bullet$ lifts $f$). Any two such liftings of $f$ are homotopic. Hence,*

$$\mathrm{Hom}_{\mathbf{K}(\mathcal{A})}(I_A^\bullet, I_B^\bullet) \simeq \mathrm{Hom}_{\mathcal{A}}(A, B).$$

*Proof.* First we construct one such lifting $f^\bullet$. Since $A \to I_A^0$ is injective, the map $d_B \circ f : A \to I_B^0$ lifts to some $f^0 : I_A^0 \to I_B^0$ by injectivity of $I_B^0$. Suppose by induction that we have defined $f^k$ for $k \leq n$ such that $d_B \circ f^k = f^{k+1} \circ d_A$ for all $k < n$. Note that $d_B \circ f^n$ sends $\ker d_A^n$ to $\mathrm{im}(d_B \circ f^n \circ d_A^{n-1}) = \mathrm{im}(d_B \circ d_B \circ f^{n-1}) = 0$. Hence $d_B \circ f^n$ descends to a map $d_A(I_A^n) \to I_B^{n+1}$, and by injectivity of $I_B^{n+1}$ this lifts to a map $f^{n+1} : I_A^{n+1} \to I_B^{n+1}$. By construction this map satisfies $f^{n+1} \circ d_A = d_B \circ f^n$. Indeed, as $i_B \circ f = f^0 \circ i_A$, we have $H^0(f^\bullet) = f$.

To show two such liftings $f_1^\bullet$ and $f_2^\bullet$ are homotopic, we can equivalently show that $f_1^\bullet - f_2^\bullet$ is homotopic to the zero map $I_A^\bullet \to I_B^\bullet$. Note that $f_1^\bullet - f_2^\bullet$ lifts the zero map $A \to B$, so it suffices to take $f = 0$ and show that the obtained map $f^\bullet$ is of the form $f^n = d_B \circ h^n + h^{n+1} \circ d_A$.

Since $f^0(A) = 0$, the map $f^0$ descends to a map $d(I_A^0) \to I_B^0$, which lifts to a map $h^1 : I_A^1 \to I_B^0$ by injectivity of $I_B^0$. Again by induction, assume that we have maps $h^k : I_A^k \to I_B^{k-1}$ for all $k < n$, such that $f^k = d_B \circ h^k + h^{k+1} \circ d_A$ for all $k < n$. Then, note that $f^n - d_B h^n$ sends $\ker d_A^n$ to

$$\operatorname{im}((f^n - d_B \circ h^n) \circ d_A) = \operatorname{im}(d_B \circ f^{n-1} - d_B \circ (f^{n-1} - d_B \circ h^{n-1})) = 0,$$

so $f^n - d_B \circ h^n$ descends to a map $d(I_A^n) \to I_B^n$ and by injectivity of $I_B^n$ lifts to a map $h^{n+1} : I_A^{n+1} \to I_B^n$, which satisfies $f^n = d_B \circ h^n + h^{n+1} \circ d_A$ by construction. $\square$

**Definition 1.4.** A chain map $f^\bullet : A^\bullet \to B^\bullet$ is called a *quasi-isomorphism* if the induced maps $H^i(f) : H^i(A) \to H^i(B)$ are isomorphisms for all $i$.

**Example 1.5.** A quasi-isomorphisms is generally not invertible. Take for example the map of complexes:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "\cdot 2"] \arrow[d] & \mathbb{Z} \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

Also there are complexes which have the same homology, but are not quasi-isomorphic. For example,

$$0 \to \mathbb{C}[x,y] \oplus \mathbb{C}[x,y] \xrightarrow{\varphi} \mathbb{C}[x,y] \to 0 \qquad \text{and} \qquad 0 \to \mathbb{C}[x,y] \xrightarrow{0} \mathbb{C} \to 0$$

with $\varphi(f,g) = xf + yg$.

**Definition 1.6.** The *derived category* $\mathbf{D}(\mathcal{A})$ is obtained from $\mathbf{K}(\mathcal{A})$ by 'pretending' the quasi-isomorphisms are isomorphisms. Concretely, a morphism from $A^\bullet$ to $B^\bullet$ is given by a roof

$$\begin{tikzcd}
 & C^\bullet \arrow[dl, "f"'] \arrow[dr, "g"] & \\
A^\bullet \arrow[rr, dashed] & & B^\bullet,
\end{tikzcd}$$

where $f$ is a quasi-isomorphism, and we think of this roof as $g \circ f^{-1}$ (even though $f^{-1}$ might not exist). More formally, we *localize* the category $\mathbf{K}(\mathcal{A})$.

## 2 Exact Triangles

Unfortunately, when passing to $\mathbf{K}(\mathcal{A})$, we lose the notion of kernels, cokernels and exact sequences, i.e. $\mathbf{K}(\mathcal{A})$ is not an abelian category. However, thanks to Jean-Louis Verdier we do have the notion of *exact triangles*.

**Definition 2.1.** An *exact triangle* in $\mathbf{K}(\mathcal{A})$ is a diagram

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \xrightarrow{h} A^\bullet[1]$$

which is isomorphic to such a diagram with $C^i = A^{i+1} \oplus B^i$ and

$$d_C^i = \begin{pmatrix} -d_A^{i+1} & 0 \\ f^{i+1} & d_B \end{pmatrix}$$

and with $g$ and $h$ the obvious maps (that is, $C^\bullet$ is the *cone* of $f$). This makes $\mathbf{K}(\mathcal{A})$ into a *triangulated category*, and one can show that this makes $\mathbf{D}(\mathcal{A})$ into one as well.

**Example 2.2.** Let $0 \to A \to B \to C \to 0$ be a short exact sequence in $\mathcal{A}$. We obtain a morphism $h : C \to A[1]$ in $\mathbf{D}(\mathcal{A})$ from the following complexes:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] & C \arrow[r] & \cdots & & C \arrow[dd, "h"] \\
\cdots \arrow[r] & A \arrow[r, "f"] \arrow[u] \arrow[d, "\mathrm{id}"'] & B \arrow[r] \arrow[u, "g"'] \arrow[d] & \cdots & & \\
\cdots \arrow[r] & A \arrow[r] & 0 \arrow[r] & \cdots & & A[1],
\end{tikzcd}$$

since the upper map of complexes is a quasi-isomorphism. This diagram also immediately shows that

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} A[1]$$

is an exact triangle in $\mathbf{D}(\mathcal{A})$, because the middle row is the cone of $f$.

**Example 2.3.** Let $A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \xrightarrow{h} A^\bullet[1]$ be an exact triangle in $\mathbf{D}(\mathcal{A})$. By definition, it is isomorphic (in $\mathbf{D}(\mathcal{A})$) to $A^\bullet \xrightarrow{f} B^\bullet \to \mathrm{Cone}(f) \to A^\bullet[1]$. In particular, $0 \to B^\bullet \to \mathrm{Cone}(f) \to A^\bullet[1] \to 0$ is a short exact sequence of complexes, so we obtain a long exact sequence

$$\cdots \to H^i(A^\bullet) \to H^i(B^\bullet) \to H^i(\mathrm{Cone}(f)) \to H^{i+1}(A^\bullet) \to \cdots$$

or equivalently, since the triangles are quasi-isomorphic (as complexes), the long exact sequence

$$\cdots \to H^i(A^\bullet) \to H^i(B^\bullet) \to H^i(C^\bullet) \to H^{i+1}(A^\bullet) \to \cdots$$

# 3 Derived Functors

**Definition 3.1.** A *bounded complex* is a complex $A^\bullet$ such that $H^i(A^\bullet)$ for $i \ll 0$ and $i \gg 0$. We denote the full subcategory of these in $\mathbf{K}(\mathcal{A})$ by $\mathbf{K}^b(\mathcal{A})$, and the resulting localization is denoted $\mathbf{D}^b(\mathcal{A})$.

Similarly, $\mathbf{K}^+(\mathcal{A})$ and $\mathbf{K}^+(\mathcal{A})$ denote the full subcategories of complexes that are zero for $i \ll 0$ and $i \gg 0$, respectively. The resulting localizations are $\mathbf{D}^+(\mathcal{A})$ and $\mathbf{D}^-(\mathcal{A})$.

**Proposition 3.2.** *Let $f : A^\bullet \to B^\bullet$ be a quasi-isomorphism in $\mathbf{K}^b(\mathcal{A})$ of bounded below complexes of injective objects of $\mathcal{A}$. Then $f$ is invertible, i.e. there exists a chain map $g : B^\bullet \to A^\bullet$ such that $g \circ f$ and $f \circ g$ are homotopic to $\mathrm{id}_{A^\bullet}$ and $\mathrm{id}_{B^\bullet}$, respectively.*

*Proof.* (Write proof) □

**Proposition 3.3.** *Suppose $\mathcal{A}$ has enough injectives. Then every bounded below complex of $\mathbf{K}^b(\mathcal{A})$ is quasi-isomorphic to a complex of injectives.*

*Proof.* (Write proof) (Proposition can be extended to $\mathbf{K}^+(\mathcal{A})$) □

**Corollary 3.4.** *The composition of the following sequence of functors is an equivalence of categories:*

$$\mathbf{K}^b(\mathrm{inj}(\mathcal{A})) \xrightarrow{i} \mathbf{K}^b(\mathcal{A}) \xrightarrow{Q} \mathbf{D}^b(\mathcal{A}),$$

*where $i$ is the inclusion, and $Q$ the localization functor.*

**Proposition 3.5.** *Assume that $F : \mathcal{A} \to \mathcal{B}$ is a left exact functor between abelian categories, and suppose that $\mathcal{A}$ has enough injectives. Then $F$ induces an exact functor*

$$\bar{F} : \mathbf{K}^+(\mathrm{inj}(\mathcal{A})) \to \mathbf{K}^+(\mathcal{B})$$

*obtained by directly applying $F$ to a complex of injectives of $\mathcal{A}$.*

*Proof.* (Write proof) □

**Definition 3.6.** Fixing an inverse $Q^{-1}$, we define the functor $\mathbf{R}F$ to be the composition

$$\mathbf{D}^b(\mathcal{A}) \xrightarrow{Q^{-1}} \mathbf{K}^b(\mathrm{inj}(\mathcal{A})) \xrightarrow{\bar{F}} \mathbf{K}^b(\mathcal{B}) \xrightarrow{Q} \mathbf{D}^b(\mathcal{B}).$$

For a single object $A$ of $\mathcal{A}$, what $Q^{-1}$ does is take an injective resolution $I^\bullet$ of $A$, then $\bar{F}$ simply applies $F$ to this resolution, and we declare the result to be $\mathbf{R}F(A)$. If we were to take homology, we would find

$$H^i(\mathbf{R}F(A)) = H^i(F(I^\bullet)) = \mathbf{R}^i F(A),$$

the $i$-th right derived functor of $F$. What we have gained is that we can now do this, not just for objects, but for arbitrary complexes.

**Example 3.7.** Assume $\mathcal{A}$ has enough injectives, then

$$\mathrm{Hom}_{\mathbf{D}^b(\mathcal{A})}(A, B[i]) \simeq \mathrm{Ext}^i_{\mathcal{A}}(A, B),$$

and in particular there are no morphisms $A \to B[i]$ for $i < 0$.

**Remark 3.8.** Completely analogous to the above, one can define a functor $\mathbf{L}F : \mathbf{D}^b(\mathcal{A}) \to \mathbf{D}^b(\mathcal{B})$ for any right exact functor $F : \mathcal{A} \to \mathcal{B}$, provided that $\mathcal{A}$ has enough projective objects.

**Example 3.9.** A list of right and left derived functors relating to sheaves on some scheme $X$ include

- sheaf cohomology: $\mathbf{R}\Gamma(X, -)$, whose cohomology groups are $H^\bullet(X, -)$,
- Hom-sheaves: $\mathbf{R}\underline{\mathrm{Hom}}(-, -)$ (right derived in the second variable), whose cohomology groups are $\underline{\mathrm{Ext}}^\bullet_X(-, -)$,
- $\mathcal{O}_X$-module morphisms: $\mathbf{R}\,\mathrm{Hom}(-, -)$, whose cohomology groups are $\mathrm{Ext}^\bullet_X(-, -)$,
- tensor product of sheafs $\mathbf{L}(- \otimes -) = - \overset{\mathbf{L}}{\otimes} -$ with homology groups $\mathrm{Tor}^X_\bullet(-, -)$,
- $\mathbf{R}f_*$ and $\mathbf{L}f^*$ for a morphism of schemes $f : X \to Y$.

# 4 Triangulated Terminology

**Definition 4.1.** Let $A_0, A_1, \ldots, A_n$ be a sequence of objects in a triangulated category. We write

$$\langle A_0, A_1, \ldots, A_n \rangle$$

for the smallest full subcategory that contains these objects and is closed under shifts and taking cones. We say the triangulated category is *generated* by these objects if it is equivalent to this resulting subcategory.

**Definition 4.2.** Say we work over a base field $k$. A sequence of objects $A_1, \ldots, A_n$ in a triangulated category is called *exceptional* if

$$\mathrm{Ext}^i(A_p, A_q) = 0 \text{ for all } i, \text{ if } p > q,$$

and

$$\mathrm{Ext}^i(A_p, A_p) = \begin{cases} 0 & \text{if } i > 0, \\ k & \text{if } i = 0. \end{cases}$$

**Example 4.3.** Given an exceptional object $E$, the subcategory $\langle E \rangle$ generated by $E$ is equivalent to $\mathbf{D}^{\mathrm{b}}(\mathbf{FinVect}_k)$, the bounded derived category of finite vector spaces over $k$. For example, every object in $\langle E \rangle$ is isomorphic to a direct sum of shifts of $E$.

**Remark 4.4.** The reason exceptional sequences are useful is because a derived category constructed using one can be thought of as being pieced together in the simplest possible way: the category generated by any single object in the exceptional sequence is equivalent to the derived category of vector spaces, and we build the larger category by adding these 'one dimensional subspaces' one after another. Furthermore, at each step we have

$$\mathrm{Hom}(\langle A_1, \ldots, A_i \rangle, \langle A_j \rangle) = 0 \text{ for } i < j.$$

**Definition 4.5.** Let $\mathcal{T}$ be a triangulated category. A *right admissible* (resp. *left admissible*) subcategory of $\mathcal{T}$ is a strictly full triangulated subcategory $\mathcal{A}$ such that the inclusion functor $i : \mathcal{A} \to \mathcal{T}$ has a right (resp. left) adjoint.

This is equivalent to saying that for every $X$ in $\mathcal{T}$ there exists a distinguised triangle $A \to X \to B \to A[1]$ with $A$ in $\mathcal{A}$ and $B$ in $\mathcal{A}^{\perp}$ (resp. $B \to X \to A \to B[1]$ with $A$ in $\mathcal{A}$ and $B$ in ${}^{\perp}\mathcal{A}$).

**Definition 4.6.** Let $\mathcal{D}$ be a triangulated category. A sequence of full admissible triangulated subcategories $\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_n \subset \mathcal{D}$ is **semi-orthogonal** if

$$\mathcal{D}_i \subset \mathcal{D}_j^{\perp} \qquad \text{for all } i < j.$$

Such a sequence defines a **semi-orthogonal decomposition** of $\mathcal{D}$ if $\mathcal{D}$ is generated by the $\mathcal{D}_i$. In this case one writes

$$\mathcal{D} = \langle \mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_n \rangle.$$

# 5 The Derived Category of $\mathbb{P}^n$

We will try to 'compute' the derived category of $\mathbb{P}^n$, that is $\mathbf{D}^b_{\mathrm{coh}}(\mathbb{P}^n)$.

As our starting point, we take the *Beĭlinson resolution*, a resolution of $\mathcal{O}_\Delta$ in $\mathbb{P}^n \times \mathbb{P}^n$, given by

$$\begin{aligned} 0 \to \mathcal{O}(-n) \boxtimes \Omega^n(n) \to \mathcal{O}(-n+1) \boxtimes \Omega^{n-1}(n-1) \to \cdots \\ \cdots \to \mathcal{O}(-1) \boxtimes \Omega^1(1) \to \mathcal{O} \boxtimes \mathcal{O} \to \mathcal{O}_\Delta \to 0, \end{aligned}$$

where the $\boxtimes$ operation is defined as (think of as the derived version of the exterior product)

$$\mathcal{E} \boxtimes \mathcal{F} = \pi_X^* \mathcal{E} \overset{\mathbf{L}}{\otimes} \pi_Y^* \mathcal{F} \in \mathbf{D}^b(X \times Y),$$

for $\mathcal{E} \in \mathbf{D}^b_{\mathrm{coh}}(X)$ and $\mathcal{F} \in \mathbf{D}^b_{\mathrm{coh}}(Y)$. In particular, this gives for every $\mathcal{A} \in \mathbf{D}^b_{\mathrm{coh}}(\mathbb{P}^n)$ an exact sequence

$$\begin{aligned} 0 \to \Phi^{\mathcal{O}(-n)\boxtimes\Omega^n(n)}(\mathcal{A}) \to \Phi^{\mathcal{O}(-n+1)\boxtimes\Omega^{n-1}(n-1)}(\mathcal{A}) \to \cdots \\ \cdots \to \Phi^{\mathcal{O}(-1)\boxtimes\Omega^1(1)}(\mathcal{A}) \to \Phi^{\mathcal{O}\boxtimes\mathcal{O}}(\mathcal{A}) \to \Phi^{\mathcal{O}_\Delta}(\mathcal{A}) \to 0. \end{aligned}$$

Let $p, q, u, v$ be as in the following diagram:

$$\begin{tikzcd} \mathbb{P}^n \times \mathbb{P}^n \arrow[r, "p"] \arrow[d, "q"'] & \mathbb{P}^n \arrow[d, "v"] \\ \mathbb{P}^n \arrow[r, "u"] & \operatorname{Spec} \mathbb{C} \end{tikzcd}$$

Note that $\Phi^{\mathcal{O}_\Delta}(\mathcal{A}) \simeq \mathcal{A}$, and furthermore we have

$$\begin{aligned} \Phi^{\mathcal{O}(-i)\boxtimes\Omega^i(i)}(\mathcal{A}) &= \mathbf{R}q_*(\mathbf{L}p^*\mathcal{A} \overset{\mathbf{L}}{\otimes} (\mathcal{O}(-i) \boxtimes \Omega^i(i))) \\ &= \mathbf{R}q_*(\mathbf{L}p^*\mathcal{A} \overset{\mathbf{L}}{\otimes} q^*\mathcal{O}(-i) \overset{\mathbf{L}}{\otimes} \mathbf{L}p^*\Omega^i(i))) \\ &\simeq \mathcal{O}(-i) \overset{\mathbf{L}}{\otimes} \mathbf{R}q_*\mathbf{L}p^*(\mathcal{A} \overset{\mathbf{L}}{\otimes} \Omega^i(i))) \\ &\simeq \mathcal{O}(-i) \otimes_{\mathbb{C}} \mathbf{R}\Gamma(\mathbb{P}^n, \mathcal{A} \overset{\mathbf{L}}{\otimes} \Omega^i(i)), \end{aligned}$$

where in the third line we used the projection formula, and in the last line the natural isomorphism of functors

$$\mathbf{R}q_* \circ \mathbf{L}p^* \simeq \mathbf{L}u^* \circ \mathbf{R}v_*,$$

and the fact that $\mathbf{R}v_* = \mathbf{R}\Gamma(\mathbb{P}^n, -)$. Hence, in position $k$, this complex has $\dim \mathbf{R}^k\Gamma(\mathbb{P}^n, \mathcal{A} \overset{\mathbf{L}}{\otimes} \Omega^i(i))$ copies of $\mathcal{O}(-i)$, and thus is generated by $\mathcal{O}(-i)$. In combination with the Beĭlinson resolution, this shows that $\mathbf{D}^b_{\mathrm{coh}}(\mathbb{P}^n)$ is generated by the $\mathcal{O}(-i)$, and moreover the sequence $\langle \mathcal{O}(-n), \mathcal{O}(-n+1), \ldots, \mathcal{O}(-1), \mathcal{O} \rangle$ is an exceptional sequence, since every morphism $\mathcal{O}(-i) \to \mathcal{O}(-j)$ is zero for $i < j$.

In conclusion, we have a semi-orthogonal decomposition by an exceptional sequence

$$\mathbf{D}^b_{\mathrm{coh}}(\mathbb{P}^n) = \langle \mathcal{O}(-n), \mathcal{O}(-n+1), \ldots, \mathcal{O}(-1), \mathcal{O} \rangle.$$

# 6 Serre functors

**Definition 6.1.** Let $\mathcal{C}$ be a $k$-linear category (i.e. its hom-sets are $k$-vector spaces and composition is bilinear). Then an auto-equivalence functor $S : \mathcal{C} \to \mathcal{C}$ is a *Serre functor* for $\mathcal{C}$ if there is a natural bijection

$$\mathrm{Hom}_{\mathcal{C}}(A, B) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(B, SA)^{\vee}$$

for all $A, B$ in $\mathcal{C}$.

**Theorem 6.2.** *Let $\mathcal{C}$ be a triangulated, $k$-linear category. If a Serre functor exists, it is unique up to unique isomorphism of functors. It is exact (i.e. takes exact triangular to exact triangles), commutes with translation, and if $F : \mathcal{C} \to \mathcal{D}$ is a triangulated equivalence, then there is a canonical isomorphism*

$$F \circ S_{\mathcal{C}} \simeq S_{\mathcal{D}} \circ F.$$

The great thing about Serre functors is that it allows one to convert from a left adjoint to a right adjoint and vice versa.

**Theorem 6.3.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between $k$-linear categories $\mathcal{C}$ and $\mathcal{D}$ admitting Serre functors $S_{\mathcal{C}}$ and $S_{\mathcal{D}}$. Assume $F$ has a left adjoint $G : \mathcal{D} \to \mathcal{C}$, i.e. $G \dashv F$. Then*

$$H = S_{\mathcal{C}} \circ G \circ S_{\mathcal{D}}^{-1} : \mathcal{D} \to \mathcal{C}$$

*is a right adjoint to $F$.*

*Proof.* We have

$$\mathrm{Hom}_{\mathcal{D}}(Fx, y) \simeq \mathrm{Hom}_{\mathcal{D}}(S_{\mathcal{D}}^{-1}y, Fx)^{\vee} \simeq \mathrm{Hom}_{\mathcal{C}}(GS_{\mathcal{D}}^{-1}y, x)^{\vee} \simeq \mathrm{Hom}_{\mathcal{C}}(x, S_{\mathcal{C}}GS_{\mathcal{D}}^{-1}y) = \mathrm{Hom}_{\mathcal{C}}(x, Hy),$$

where every step is a natural bijection. □

**Example 6.4.** Let $X$ be a smooth projective variety of dimension $n$. Then Serre duality gives natural isomorphisms

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}) \simeq H^i(X, \mathcal{F}^{\vee} \otimes \mathcal{G}) \simeq H^{n-i}(X, \mathcal{F} \otimes \mathcal{G}^{\vee} \otimes \omega_X) \simeq \mathrm{Ext}_X^{n-1}(\mathcal{G}, \mathcal{F} \otimes \omega_X)^{\vee},$$

for coherent sheaves $\mathcal{F}, \mathcal{G}$ on $X$. In the derived category $\mathbf{D}^b_{\mathrm{coh}}(X)$ this can be rewritten as

$$\mathrm{Hom}_{\mathbf{D}^b(X)}(\mathcal{F}, \mathcal{G}[i]) \simeq \mathrm{Hom}_{\mathbf{D}^b(X)}(\mathcal{G}[i], \mathcal{F} \otimes \omega_X[n])^{\vee}$$

This shows that $\mathbf{D}^b_{\mathrm{coh}}(X)$ admits a Serre functor given by

$$S_X : \mathbf{D}^b_{\mathrm{coh}}(X) \to \mathbf{D}^b_{\mathrm{coh}}(X), \qquad \mathcal{F} \mapsto \mathcal{F} \otimes \omega_X[\dim X].$$

Historically, one of the motivating examples was to find a right adjoint to the push-forward functor $f_*$ for $f : X \to Y$ a morphism of schemes. Generally this is impossible (since $f_*$ is only left exact), but for a projective morphism $f : X \to Y$ we do have a left adjoint, $f^* \dashv f_*$. When passing to derived categories, we replace $f_*$ by the right-derived (exact) functor $\mathbf{R}f_*$, and using Theorem 6.3 we find:

**Corollary 6.5.** *Let $f : X \to Y$ be a morphism of smooth projective schemes. Then a right adjoint to $\mathbf{R}f_*$ exists, and is given by*

$$f^! : \mathbf{D}^b_{\mathrm{coh}}(Y) \to \mathbf{D}^b_{\mathrm{coh}}(X), \qquad \mathcal{F} \mapsto \mathbf{L}f^*(\mathcal{F} \otimes \omega_Y^{-1}[-\dim Y]) \otimes \omega_X[\dim X].$$

# 7 From derived category to schemes

One may wonder how much information the derived category of a scheme contains about the scheme. Suppose that we have an equivalence $\mathbf{D}^b_{\mathrm{coh}}(X) \simeq \mathbf{D}^b_{\mathrm{coh}}(Y)$. In general, we do not have $X \simeq Y$, but there are counterexamples.

**Theorem 7.1** (Bondal-Orlov)**.** *Let $X$ be a projective variety such that $\omega_X$ is either ample or anti-ample, and let $Y$ be any projective variety. If $\mathbf{D}^b_{\mathrm{coh}}(X) \simeq \mathbf{D}^b_{\mathrm{coh}}(Y)$ as triangulated categories, then $X \simeq Y$.*

This can be proven in some steps.

1. Identify shifts of point sheaves, $\mathcal{O}_x[i]$, categorically. This allows for a bijection of closed points from $X$ to $Y$.
2. Identify line bundles categorically.
3. Do some magic with these line bundles to conclude continuous map of topological spaces.
4. Do some more magic to have an isomorphism of pluricanonical rings. By ample/anti-ampleness we have $X \simeq Y$.

# 8 Fourier-Mukai transforms

**Definition 8.1.** Let $X$ and $Y$ be smooth projective varieties. For any $\mathcal{E}$ in $\mathbf{D}^b_{\mathrm{coh}}(X \times Y)$, the *integral transform with kernel* $\mathcal{E}$ is the functor given by

$$\Phi^{\mathcal{E}}_{X\to Y} : \mathbf{D}^b_{\mathrm{coh}}(X) \to \mathbf{D}^b_{\mathrm{coh}}(Y), \qquad \Phi^{\mathcal{E}}_{X\to Y}(-) = \mathbf{R}\pi_{Y,*}\left(\pi_X^*(-) \overset{\mathbf{L}}{\otimes} \mathcal{E}\right).$$

The composition of integral transforms is again an integral transform: when $\mathcal{E} \in \mathbf{D}^b_{\mathrm{coh}}(X \times Y)$ and $\mathcal{F} \in \mathbf{D}^b_{\mathrm{coh}}(Y \times Z)$, then

$$\Phi^{\mathcal{F}}_{Y\to Z} \circ \Phi^{\mathcal{E}}_{X\to Y} = \Phi^{\mathcal{G}}_{X\to Z} \qquad \text{with} \qquad \mathcal{G} = \mathbf{R}\pi_{XZ,*}\left(\pi^*_{XY}\mathcal{E} \overset{\mathbf{L}}{\otimes} \pi^*_{YZ}\mathcal{F}\right).$$

The functor $\Phi^{\mathcal{E}}_{X\to Y}$ always has both a left and right adjoint, respectively given by

$$\Phi^{\mathcal{E}^\vee \otimes \omega_Y[\dim Y]}_{Y\to X} \qquad \text{and} \qquad \Phi^{\mathcal{E}^\vee \otimes \omega_X[\dim X]}_{Y\to X}$$

where $\mathcal{E}^\vee = \mathbf{R}\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{O}_{X\times Y})$.

**Example 8.2.** Let $f : X \to Y$ be a morphism, and let $\Gamma_f \subset X \times Y$ be its graph. Using the sheaf $\mathcal{O}_{\Gamma_f}$ as a kernel induces the integral transforms

$$\mathbf{R}f_* = \Phi_{X\to Y}^{\mathcal{O}_{\Gamma_f}} \qquad \text{and} \qquad \mathbf{L}f^* = \Phi_{Y\to X}^{\mathcal{O}_{\Gamma_f}}.$$

Furthermore, for any $\mathcal{E} \in \mathbf{D}^b_{\mathrm{coh}}(X)$ we have

$$\Phi_{X\to X}^{\Delta_*\mathcal{E}}(-) = - \overset{\mathbf{L}}{\otimes} \mathcal{E} : \mathbf{D}^b_{\mathrm{coh}}(X) \to \mathbf{D}^b_{\mathrm{coh}}(X).$$

In fact, the only exact functors $\mathbf{D}^b_{\mathrm{coh}}(X) \to \mathbf{D}^b_{\mathrm{coh}}(Y)$ anybody knows are integral transforms.

**Definition 8.3.** An integral transform which is an equivalence is called a *Fourier–Mukai transform.* The reason for this name is that Mukai came up with it, and it has similarities with with the Fourier transform.

**Theorem 8.4** (Mukai, Bondal-Orlov, Bridgeland)**.** *Let $X, Y$ be smooth projective varieties over $k$, and let $\Phi : \mathbf{D}^b_{\mathrm{coh}}(X) \to \mathbf{D}^b_{\mathrm{coh}}(Y)$ be an integral transform. Write $\mathcal{P}_x = \Phi(\mathcal{O}_x)$. Then $\Phi$ is fully faithful if and only if*

$$\mathrm{Hom}_{\mathbf{D}^b_{\mathrm{coh}}(X)}(\mathcal{P}_x, \mathcal{P}_y[i]) = \begin{cases} 0 & \text{if } x \neq y \text{ or } i \notin [0, \dim X], \\ k & \text{if } x = y \text{ and } i = 0. \end{cases}$$

*for all $x, y \in X$ and $i \in \mathbb{Z}$. Furthermore, $\Phi$ is an equivalence if and only if*

$$\mathcal{P}_x \otimes \omega_Y \simeq \mathcal{P}_x$$

*for all $x \in X$.*

# 9 Hochschild cohomology

Derived category is a bit involved, we would like to have invariants to compare the derived category of different schemes for example.

**Definition 9.1.** Let $X$ be a smooth projective variety. The *Hochschild cohomology* of $X$ is defined to be

$$HH^*(X) = \mathrm{Ext}^*_{X\times X}(\mathcal{O}_\Delta, \mathcal{O}_\Delta),$$

and the *Hochschild homology* of $X$ to be

$$HH^*(X) = \mathrm{Ext}^{-*}_{X\times X}(\Delta_!\mathcal{O}_X, \mathcal{O}_\Delta),$$

where $\Delta_! : \mathbf{D}^b_{\mathrm{coh}}(X) \to \mathbf{D}^b_{\mathrm{coh}}(X \times X)$ is the left adjoint of $\mathbf{L}\Delta^*$.

We have a more explicit when $X$ is nice.

**Theorem 9.2** (Hochschild–Kostant–Rosenberg)**.** *Let $X$ be a smooth, quasi-projective variety over $\mathbb{C}$. Then*

$$HH^i(X) \simeq \bigoplus_{p+q=i} H^p\left(X, \wedge^q T_X\right),$$
$$HH_i(X) \simeq \bigoplus_{q-p=i} H^p\left(X, \Omega^q_X\right).$$

Note that we have

$$HH^i(X) = \mathrm{Ext}^i_{X\times X}(\mathcal{O}_\Delta, \mathcal{O}_\Delta) = \mathrm{Hom}_{\mathbf{D}^b_{\mathrm{coh}}(X\times X)}(\mathcal{O}_\Delta, \mathcal{O}_\Delta[i]).$$

Observe that $\mathcal{O}_\Delta$ as a kernel corresponds to the identity functor, and $\mathcal{O}_\Delta[i]$ to the 'shift by $i$' functor. Then it is easy to see that a morphism $\mathcal{O}_\Delta \to \mathcal{O}_\Delta[i]$ (in $\mathbf{D}^b_{\mathrm{coh}}(X \times X)$) gives a map

$$\mathcal{F} = \Phi^{\mathcal{O}_\Delta}_{X\to X}(\mathcal{F}) \to \Phi^{\mathcal{O}_\Delta[i]}_{X\to X}(\mathcal{F}) = \mathcal{F}[i],$$

for every $\mathcal{F}$ in $\mathbf{D}^b_{\mathrm{coh}}(X)$, and these maps form a natural transformation $\mathrm{id} \Rightarrow [i]$.

Now unfortunately, we cannot define the Hochschild (co)homology completely intrinsic to $\mathbf{D}^b_{\mathrm{coh}}(X)$, but it can be shown that if $\mathbf{D}^b_{\mathrm{coh}}(X)$ and $\mathbf{D}^b_{\mathrm{coh}}(Y)$ are equivalent, then their Hochschild (co)homology agree!

Furthermore, Hochschild (co)homology is functorial with respect to integral transforms. That is, for every integral transform $\Phi : \mathbf{D}^b_{\mathrm{coh}}(X) \to \mathbf{D}^b_{\mathrm{coh}}(Y)$ there is an associated map of graded vector spaces $\Phi_* : HH_*(X) \to HH_*(Y)$ such that $(\Phi \circ \Psi)_* = \Phi_* \circ \Psi_*$ and $\mathrm{id}_* = \mathrm{id}$.